

Why Two Years?

Instructor Discussion Notes

| Quiz 2 day

| 12 minutes

| No handout, no laptops

Placement and purpose

Runs after students have submitted Lab 7 and immediately before the Pennypack Creek project. Nothing new is introduced from scratch — the whole segment reinterprets a statistic students have already computed. That makes it survivable in the ten to fifteen minutes left after a quiz.

The one idea: the test statistic students wrote in Lab 7 Part 2 is one slice of Mann-Kendall. They computed increases minus decreases at a lag of two years. Mann-Kendall adds up that same quantity at **every** lag. Students discover that the “2” was never justified, that removing it costs them nothing, and that summing over all lags creates a brand-new trap that Lab 7’s null simulation was already falling into.

Why it earns the slot before Pennypack. Mann-Kendall and its Seasonal Kendall extension are the standard trend tests in water-quality monitoring. Students are about to analyze exactly the kind of data these tests were built for. This is the handoff.

Timing

2 min Setup: recall the Lab 7 statistic

3 min Question 1 — commit before reveal

2 min The reveal: it’s one slice of a famous test

3 min Question 2 — the transitivity trap

2 min Question 3 — what is still broken

Setup (2 min)

Put the Lab 7 statistic on the board without comment:

```
def changes(array, years = 2):  
    "Increases minus decreases after two years."
```

Remind them what they concluded: far more increases than decreases across all countries and years, a p-value indistinguishable from zero, null rejected, planet warming. Then say only this: “I want to look at one character in that function.”

Circle the 2.

Question 1 — Commit before reveal (3 min)

Why two years? Write down on a scrap of paper: (a) the lag you would have chosen, and (b) whether you think the conclusion would change if we had used it.

Insist on the writing. Ten seconds of silence is enough. Then poll by show of hands: lag 1, lag 2, lag 5, lag 10, something bigger. The spread is the point — there is no consensus, and no one in the room can defend their own number.

What to listen for

- **“It doesn’t matter, the trend is obvious.”** True here, and worth conceding. Ask whether they would have said that before running the test, or only after.
- **“Longer lags see more warming.”** Correct and the seed of everything that follows. Write it on the board.

- “Just try them all and report the one with the smallest p-value.” Do not correct this. Write it on the board as Option A. It is the trap, and you will come back to it in the close.

The reveal (2 min)

You do not have to choose a lag. Add up the statistic at every lag:

$$S = \sum_{k=1}^{n-1} [\text{increases} - \text{decreases at lag } k] = \sum_{i < j} \text{sign}(x_j - x_i)$$

The right-hand form says: compare every year to every other year and count which direction each pair points. That is the Mann-Kendall statistic (Mann 1945; Kendall 1975).

Give them the payoff table. Simulated annual series, interannual SD 0.30°C, permutation null for both tests, $\alpha = 0.05$:

Record	Trend	Lag-2 test	Mann-Kendall
35 years	0.009 °C/yr	0.18	0.53
80 years	0.009 °C/yr	0.18	1.00
165 years	0.009 °C/yr	0.19	1.00
35 years	0.020 °C/yr	0.33	0.98
165 years	0.020 °C/yr	0.52	1.00

Let them sit with the left column. **The lag-2 test barely improves as the record grows from 35 years to 165 years.** A lag-2 comparison sees two years of signal against a full two years of noise no matter how long the record is; lengthening the record buys more comparisons, not better ones. Mann-Kendall’s distant pairs see a century of signal against the same noise.

Question 2 — The transitivity trap (3 min)

This is the heart of the segment. Set it up as a straightforward extension.

Mann-Kendall uses every pair of years, so for 165 years that’s 13,530 comparisons instead of about 163. To build the null distribution, can we simulate 13,530 independent coin flips?

Most of the room will say yes. Let them. Then ask for three years’ temperatures where $x_1 < x_2$ and $x_2 < x_3$, and ask what the third comparison must be.

The comparisons are not free. Transitivity forces them. Treating them as independent coin flips understates the variance badly:

n	Coin-flip Var	True Var	Understated
35	595	4,958	8.3×
165	13,530	503,617	37×

The true variance is $n(n-1)(2n+5)/18$, which Kendall worked out. The naive figure is $n(n-1)/2$.

The callback that makes this land

Look back at your Lab 7 simulation. What did `simulate_under_null` assume about the lag-2 comparisons?

It sampled independent Increase/Decrease draws — the same error, at a smaller lag and therefore a smaller magnitude, but the same error. Overlapping lag-2 differences chain through shared years, and country temperatures are spatially correlated on top of that. Lab 7's conclusion survives because the signal is enormous. Its stated confidence does not.

Question 3 — What Mann-Kendall does not fix (2 min)

We've fixed the arbitrary lag. Is the test now correct?

Draw the distinction explicitly, because students will otherwise merge these into one vague sense that “the old test was bad”:

Arbitrary lag choice	Fixed. Mann-Kendall uses all of them.
Dependence between observations	Not fixed. Transitivity is handled by Kendall's variance formula, but serial autocorrelation in the temperature series is not. Warm years follow warm years. Mann-Kendall is anti-conservative under positive autocorrelation, which is why hydrologists apply variance corrections or prewhitening.
Only monotone trends	Not fixed. Against a step change or a cooling-then-warming pattern, Mann-Kendall does poorly. It is not a general trend detector.

If a strong student asks whether Mann-Kendall is simply the best test: no. Against a genuinely linear trend with well-behaved noise, fitting a slope beats it. Mann-Kendall trades a little power for robustness to outliers and to non-normal data — which is exactly the trade you want in environmental monitoring, where a single instrument failure can produce an absurd value.

Close (1 min)

Return to Option A on the board — “try every lag and report the smallest p-value.”

That is not a variant of Mann-Kendall. Mann-Kendall commits to using all lags **before** seeing the data and pays for it with a variance formula that accounts for all of them. Option A looks at the data first and then selects the analysis that produced the answer it liked. Same set of numbers examined; completely different epistemic status.

The one-sentence version to leave on the board:

A defensible analysis fixes its choices before it sees the results.

Optional callback if the Silberzahn multi-analyst study has already come up: twenty-nine teams analyzing one dataset produced odds ratios spanning 0.89 to 2.93. The spread came from analytic choices across the board — which is what a room full of students choosing different lags just simulated in miniature.

If the segment runs short

Ask what “trend” would even mean for a series that rises, falls, and rises again to the same height. Mann-Kendall would report nothing. Whether that is a failure of the test or a failure of the question is a genuinely open discussion and can absorb five minutes.

If the quiz runs long

Cut Question 3. Setup, Question 1, the reveal, and Question 2 stand on their own in about eight minutes. Do not cut the close — it is the only part that generalizes beyond this one statistic.

Instructor reference

Power figures come from simulation: 4,000 replicates per cell, permutation-based critical values from 8,000 shuffles, one-sided at $\alpha = 0.05$, Gaussian noise with SD 0.30°C . These describe a **single** series; Lab 7 pools across countries, which raises power but also aggravates the dependence problem, since the dataset mixes individual countries with continental aggregates.

Sources if students want them: Mann, H.B. (1945), *Econometrica* 13, 245–259; Kendall, M.G. (1975), *Rank Correlation Methods*, 4th ed.; Hirsch, Slack & Smith (1982), *Water Resources Research* 18(1), 107–121, for the Seasonal Kendall test used in water quality work; Hamed & Rao (1998), *Journal of Hydrology* 204, 182–196, for the autocorrelation correction.